Forecasting Vegetable Prices in Kalimati Market (Nepal)

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Contents

1 Abstract

This project analyzes the historical price dynamics of vegetables traded in the Kalimati wholesale market (Nepal), with the goal of building forecasting models and comparing their predictive accuracy. Following exploratory data analysis and preprocessing, a range of time series approaches is evaluated, including linear regression with seasonal dummies (TSLM), exponential smoothing (ETS), seasonal ARIMA, and generalized additive models (GAM). Diffusion-type models, although considered, were not applied since they are designed for product adoption processes rather than cyclical agricultural prices. Model performance is assessed using standard forecast accuracy measures such as RMSE, MAE, and MAPE.

2 Introduction

Vegetable price volatility plays a crucial role in food security and household expenditure in Nepal. Kalimati Market, located in Kathmandu, is the largest wholesale market for agricultural commodities, making it an ideal case study to investigate both temporal patterns and forecasting methods. The aim of this study is to identify the most appropriate time series models for average price forecasting and compare them in terms of accuracy and robustness.

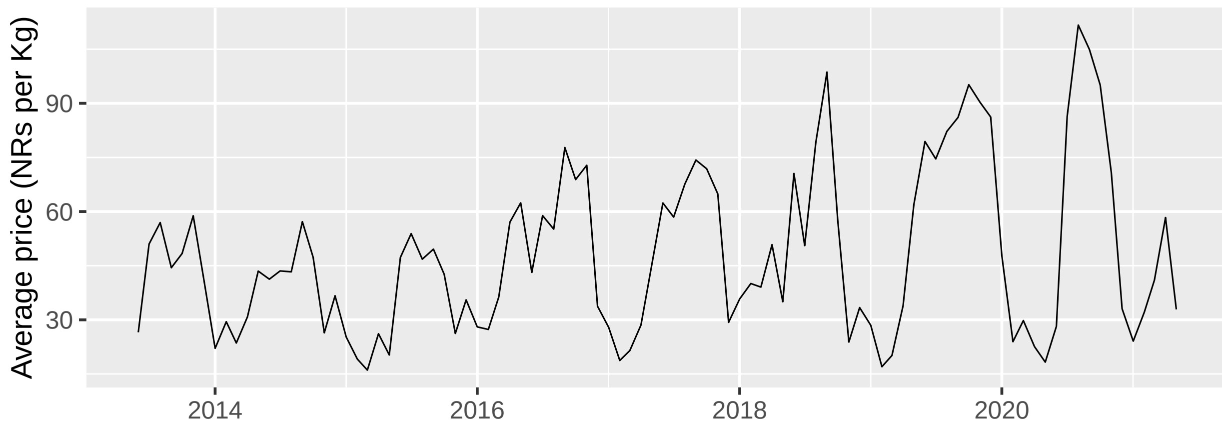
2.1 Pre Processing the Dataset

Column names and data types were standardized, and a minimal audit of the Kalimati Tarkari dataset was conducted. The sample covers the period from June 16, 2013 to May 13, 2021, with daily price records. Crucially, no missing values were detected in the average price field (avg_price) across the best-covered commodities (e.g., Ginger, Cauli Local, Cabbage, Chilli Dry, Potato Red), each with roughly 2,730–2,750 daily observations. This provides a solid foundation for time-series modeling and forecasting.

Table 1: Top-20 commodities by observation count (daily coverage)

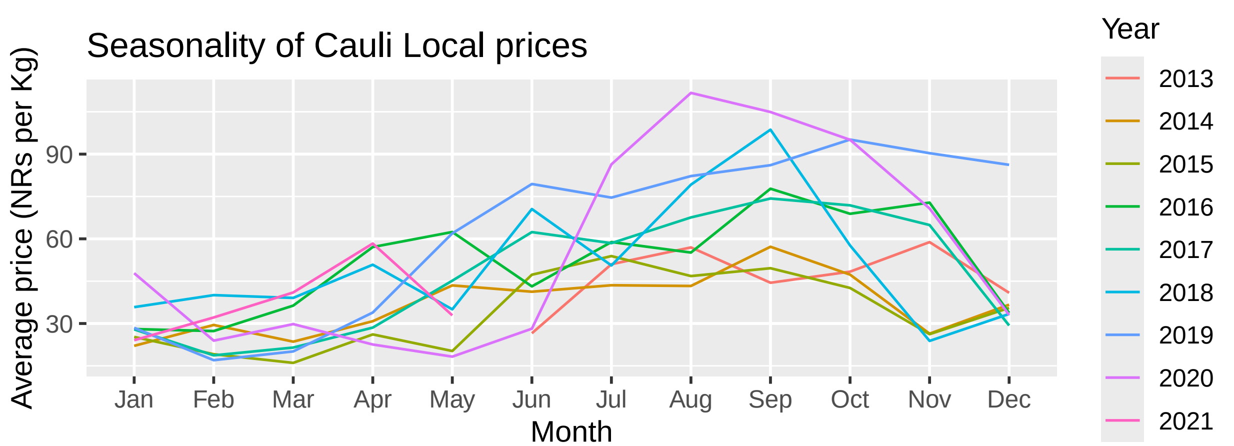
commodity	n_obs	start_date	end_date	na_avg
Ginger	2751	2013-06-16	2021-05-13	0
Cauli Local	2750	2013-06-16	2021-05-13	0
Cabbage(Local)	2749	2013-06-16	2021-05-13	0
Chilli Dry	2748	2013-06-16	2021-05-13	0
Raddish White(Local)	2747	2013-06-16	2021-05-13	0
Potato Red	2746	2013-06-16	2021-05-13	0
Bamboo Shoot	2744	2013-06-16	2021-05-13	0
Banana	2744	2013-06-16	2021-05-13	0
Brd Leaf Mustard	2742	2013-06-16	2021-05-13	0
Onion Dry (Indian)	2742	2013-06-16	2021-05-13	0
Coriander Green	2741	2013-06-16	2021-05-13	0
Tomato Small(Local)	2741	2013-06-16	2021-05-13	0
French Bean(Local)	2739	2013-06-16	2021-05-13	0
Brinjal Long	2736	2013-06-16	2021-05-13	0
Carrot(Local)	2736	2013-06-16	2021-05-13	0
Chilli Green	2735	2013-06-16	2021-05-13	0
Onion Green	2735	2013-06-16	2021-05-13	0
Spinach Leaf	2733	2013-06-16	2021-05-13	0
Mushroom(Kanya)	2730	2013-06-16	2021-05-13	0
Garlic Dry Chinese	2728	2013-06-16	2021-05-13	0

Given the daily frequency and short-term volatility, prices were aggregated at a monthly resolution. This transformation reduces noise, highlights broader trends, and facilitates the identification of seasonal patterns typical of agricultural products. The resulting series forms the basis for subsequent modeling.



The monthly line plot reveals a pronounced cyclical structure: prices fluctuate in regular patterns across the year, consistent with the seasonal nature of vegetable production and supply cycles. Peaks and troughs recur annually, and the amplitude of these cycles appears to have increased in recent years, especially after 2018. This suggests rising volatility in the cauliflower market, possibly linked to shifts in production costs, climatic shocks, or supply chain disruptions.

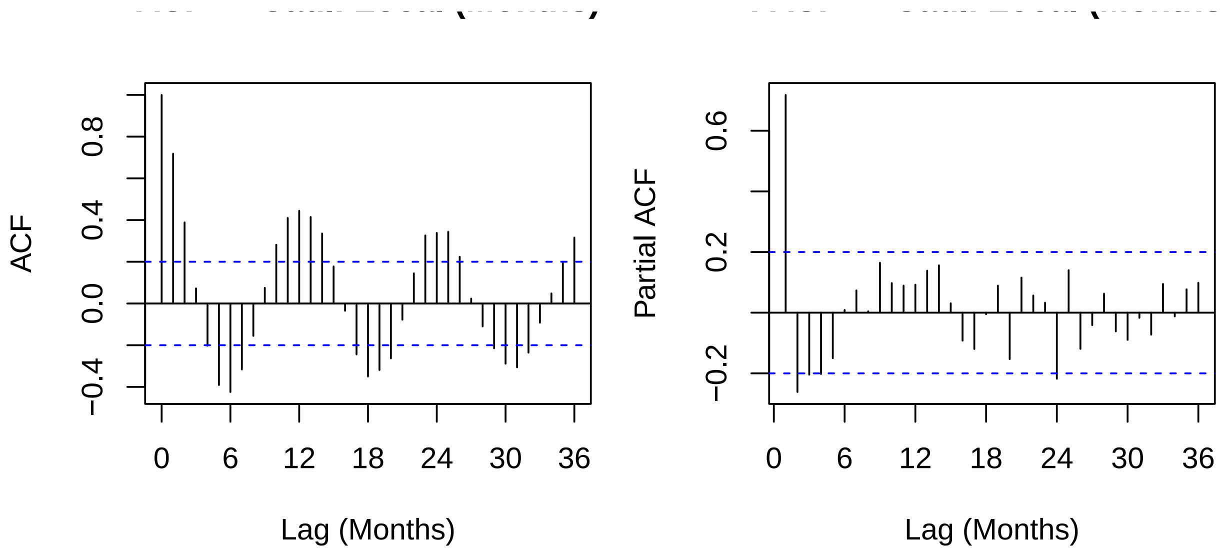
Overall, the preliminary analysis indicates that Cauli Local exhibits strong seasonality, marked inter-annual fluctuations, and increasing price variability, confirming the need for models that explicitly incorporate seasonal components and account for potential variance changes over time.



The seasonal plot further confirms this within-year cycle. Prices are generally lowest in the first quarter, increase during summer and early autumn (July–October), and decline toward the end of the year. While the overall shape is recurrent, the amplitude varies across years. In particular, 2019 and 2020 display exceptionally high peaks (above 90 NRs/kg), indicating that, in addition to regular seasonality, cauliflower prices are subject to year-specific shocks (e.g., extreme weather events or the COVID-19 period).

Implications for modeling These exploratory results highlight three key features of the cauliflower price series: -strong and regular seasonality, -inter-annual volatility and shocks, -evidence of heteroskedasticity (time-varying seasonal amplitude).

2.2 ACF and PACF Analysis



To further investigate the dependence structure of the monthly cauliflower series, the autocorrelation function (ACF) and partial autocorrelation function (PACF) were computed up to 36 lags (three years). The ACF shows strong positive correlations at short lags, particularly at lag 1, confirming a high degree of persistence in monthly prices. Moreover, pronounced spikes at multiples of 12 months provide clear evidence of annual seasonality, consistent with the agricultural production cycles of the Nepali market. The slow decay of the ACF across lags further suggests the presence of seasonal autoregressive or differencing components.

The PACF displays a dominant spike at lag 1, indicating the relevance of an AR(1) structure for the short-term dynamics, while weaker but non-negligible contributions at seasonal lags suggest the inclusion of seasonal terms. The rapid decline after the first few lags implies that only a limited autoregressive structure is required for the non-seasonal part of the model.

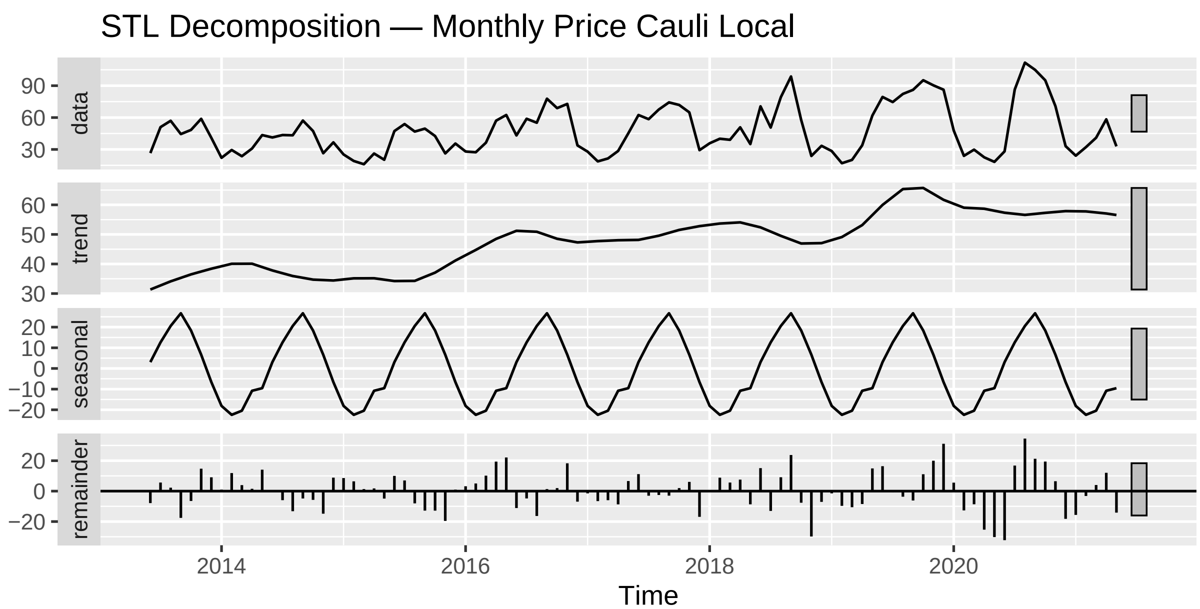
Taken together, these diagnostics confirm that seasonal ARIMA models with annual differencing and/or ETS models with multiplicative seasonality are well suited to capture the dynamics of cauliflower prices. They also reinforce the importance of considering variance-stabilizing transformations to better handle the increasing amplitude observed in the exploratory plots.

3 Models

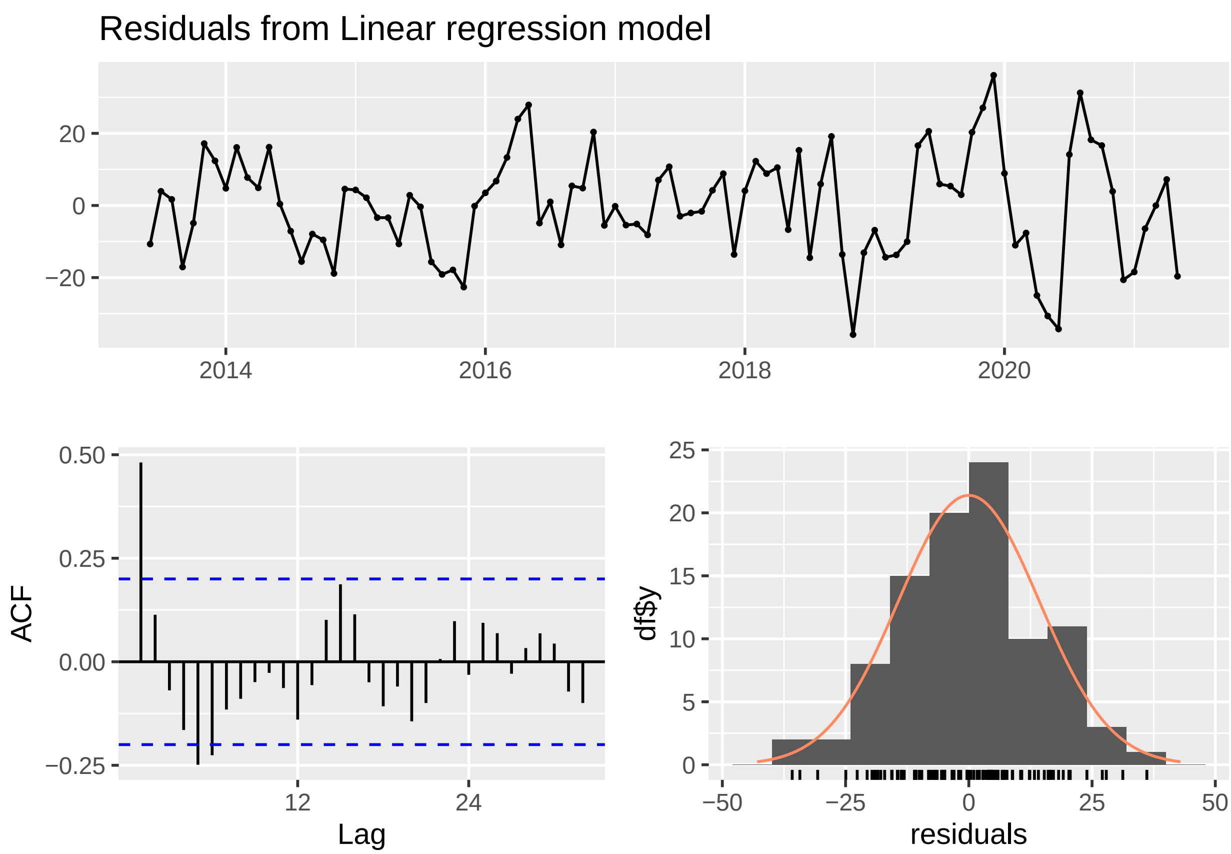
Diffusion models such as the Bass framework were not applied, since they are suited to cumulative product adoption processes, whereas agricultural prices exhibit cyclical seasonality and volatility. For this reason, time series models (ETS, SARIMA, STL+ETS, GAM) provide a more appropriate approach for forecasting in this context.

3.1 STL decomposition and baseline regression (Cauli Local)

The monthly cauliflower price series was decomposed into trend, seasonal, and remainder components using STL. The trend shows a mild long-term increase, with a slowdown around 2015–2016 and renewed growth from 2017 to 2019. The seasonal component is highly regular, reflecting the agricultural production cycle, while the remainder highlights clustered deviations, particularly in 2019–2020, suggesting episodic shocks and mild heteroskedasticity. STL is mainly exploratory, but it highlights the presence of evolving seasonality and motivates the use of flexible seasonal models.

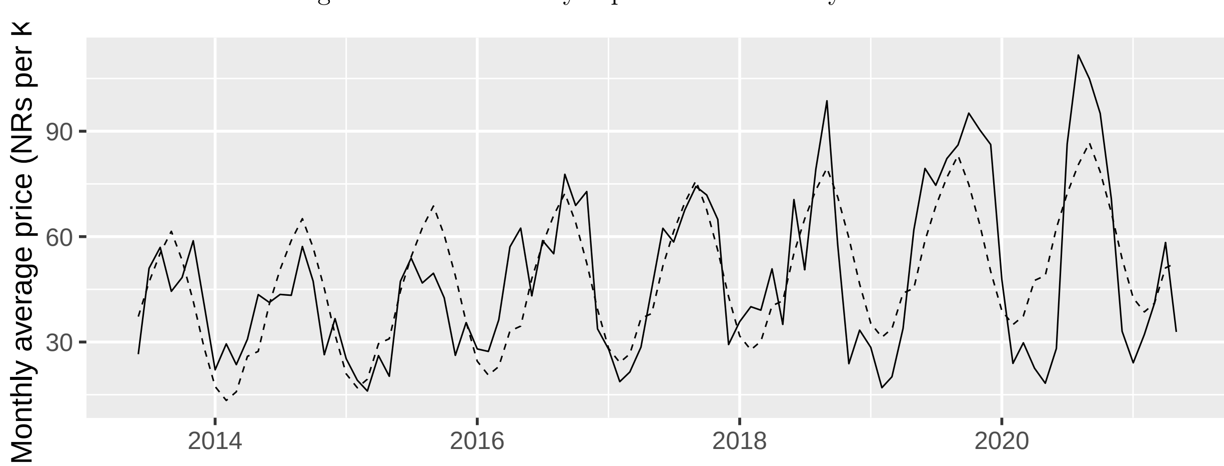


As a transparent baseline, a time series linear model with deterministic trend and monthly seasonal dummies was estimated (TSLM: xt ~ trend + season). The estimated trend is positive and statistically significant (~0.30 NRs/kg per month, ~3.6 per year), and several seasonal coefficients are significant, confirming predictable within-year patterns.




```
##
##  Breusch-Godfrey test for serial correlation of order up to 19
##
## data:  Residuals from Linear regression model
## LM test = 41.502, df = 19, p-value = 0.002068
```

The residual series oscillates around zero but exhibits clusters of positive and negative values, particularly after 2018. The ACF of residuals shows significant correlations at low lags, confirming the test result, while the histogram suggests an approximately normal distribution with slightly heavy tails. These features confirm that the baseline regression does not fully capture short-term dynamics.



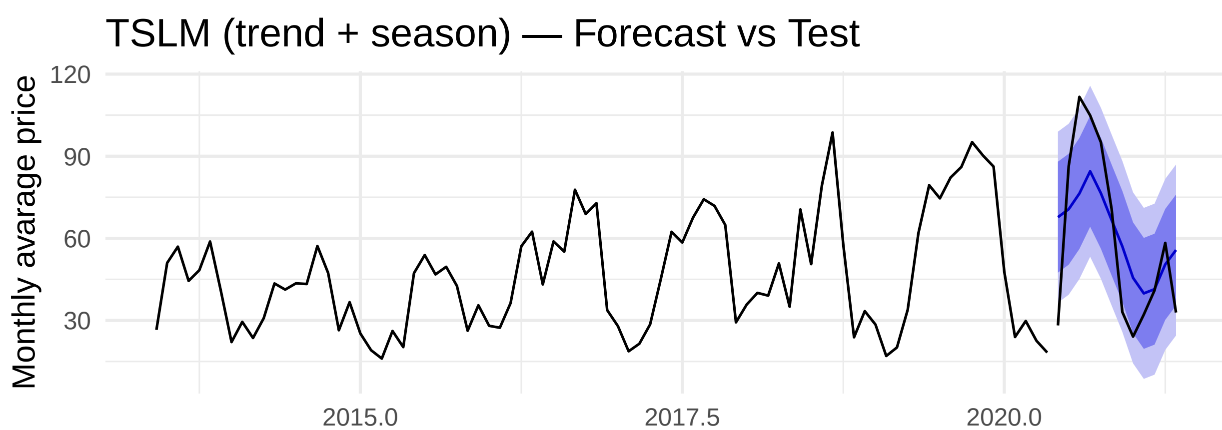
The model attains $R^2 \approx 0.61$ (adjusted $R^2 \approx 0.55$) with a residual standard error of about 15 NRs/kg.

However, residual diagnostics (Breusch–Godfrey $p = 0.002$) reveal remaining serial correlation, and the fitted series tends to underestimate peaks and overestimate troughs. These issues are consistent with additive fixed seasonality in the presence of time-varying amplitude and potential variance growth, and they motivate either a variance-stabilizing transformation (e.g., Box–Cox/log) or the use of seasonal models that natively capture serial dependence, such as ETS with multiplicative seasonality or (S)ARIMA with seasonal terms. In the next section, transformed models and a time-based train/test evaluation are introduced.

3.1.1 Hold-out evaluation of the baseline TSLM.

Using the last 12 months as a test set, the baseline TSLM achieves RMSE = 21.4, MAE = 18.2, and MAPE = 42.1%. With MASE = 1.20 (>1), the model performs worse than the seasonal naïve benchmark. Forecast intervals are wide, reflecting residual volatility and persistent autocorrelation.

Overall, while TSLM provides an interpretable starting point, its assumption of additive fixed seasonality makes it inadequate for this dataset. More flexible approaches—such as ETS with multiplicative seasonality or seasonal ARIMA—are required to capture the time-varying amplitude and volatility observed in cauliflower prices.



3.2 Regression with exogenous regressors

To explore whether cross-commodity information could improve cauliflower price forecasting, monthly price series for other major vegetables in Kalimati market—potato, cabbage, and onion—were constructed and aligned with the target series. These products were selected because they are widely consumed, have complete data coverage, and may act as substitutes or complements in household consumption, making them economically plausible predictors.

Correlation analysis shows that cauliflower prices are strongly related to cabbage (0.71), moderately to potato (0.66), and more weakly to onion (0.31). Such correlations, however, may largely reflect shared agricultural seasonality rather than causal relationships, and the high correlation between potential regressors (e.g., potato and cabbage) raises concerns of multicollinearity.

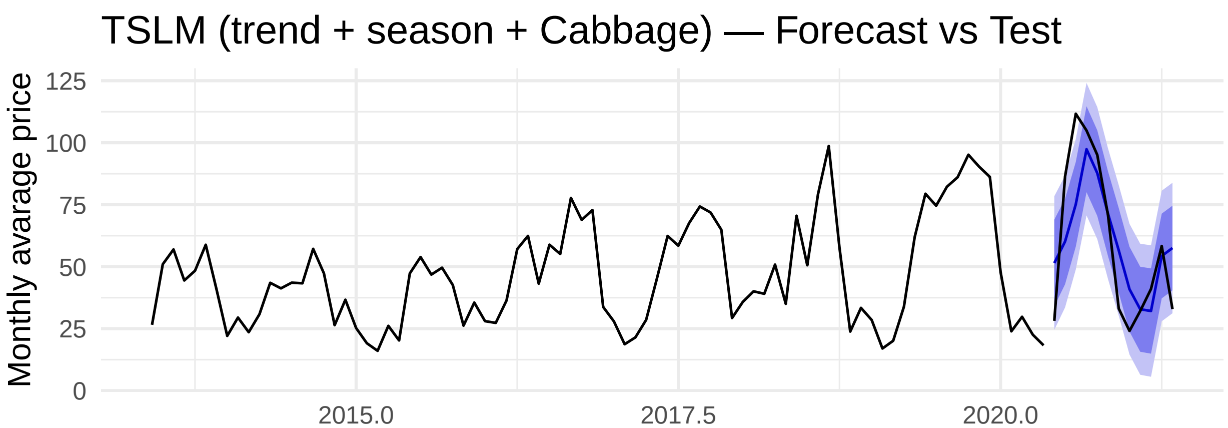
Table 2: Correlation Matrix between Average Monthly Prices

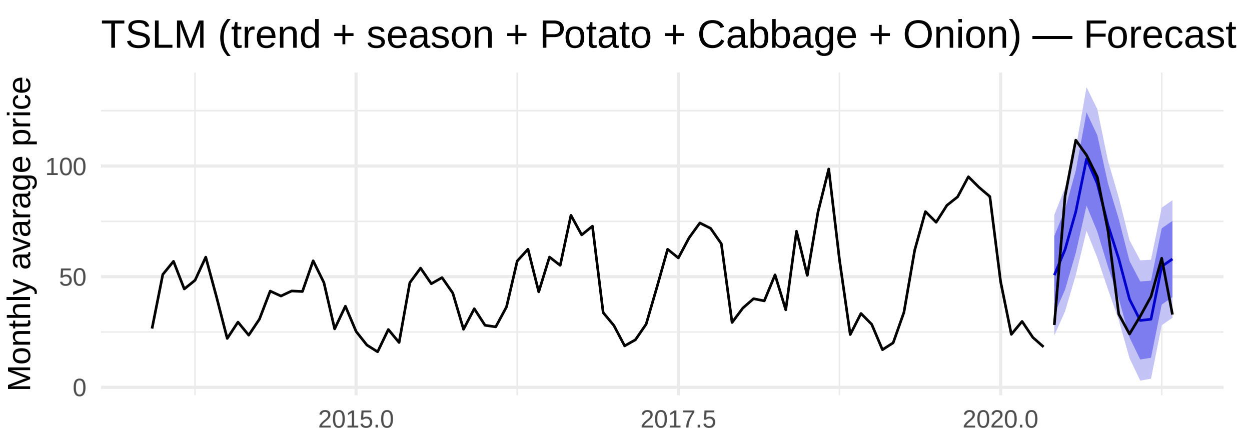
	cauli	Potato Red	Cabbage(Local)	Onion Dry (Indian)
cauli	1.00	0.66	0.71	0.31
Potato Red	0.66	1.00	0.66	0.47
Cabbage(Local)	0.71	0.66	1.00	0.62
Onion Dry (Indian)	0.31	0.47	0.62	1.00

Selection of exogenous regressors The baseline TSLM (trend + season) was first extended by including cabbage prices. The cabbage coefficient was highly significant (0.78, $p < 0.001$), confirming the strong co-movement between the two series. Out-of-sample accuracy improved compared to the baseline, with MAPE decreasing from 42% to 34%. Visually, the model tracks seasonal peaks more closely, although it still underperforms during episodes of extreme volatility.

Adding potato and onion alongside cabbage did not produce significant coefficients and only marginally reduced forecast errors (MAPE 33%). This limited gain came at the cost of increased model complexity and potential instability due to multicollinearity.

Overall, the results suggest that cabbage provides valuable explanatory information for cauliflower forecasting, while additional regressors add little value. The improvement achieved by including cabbage is real but modest, and the model remains sensitive to structural shocks and heteroskedasticity. Moreover, because this regression framework does not account for autocorrelation in residuals, its forecasts may be unstable in practice. For these reasons, more robust seasonal models (ETS, SARIMA, SARIMAX) are adopted, as they can incorporate both seasonality and exogenous information within a dynamic time-series structure.



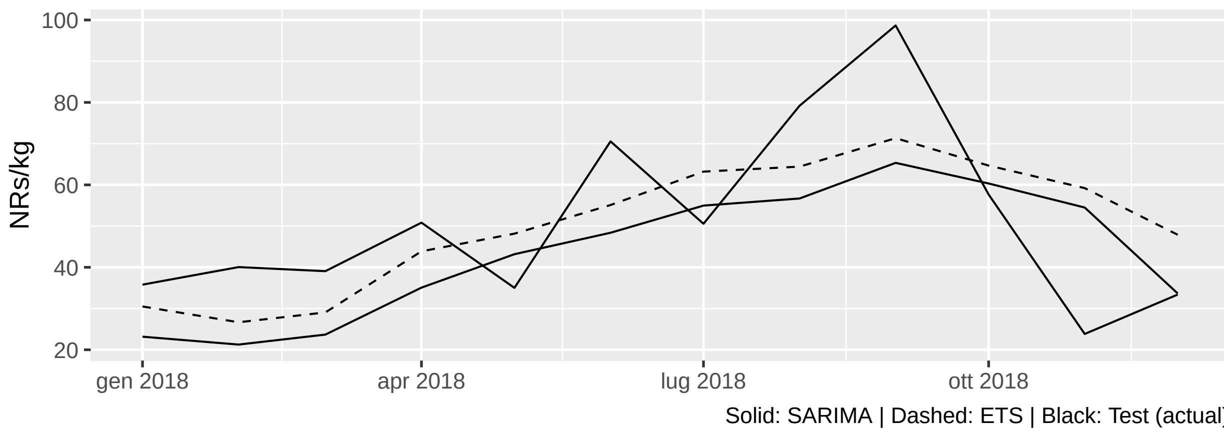


3.3 Seasonal ARIMA

Given the exceptional volatility in 2020–2021, the hold-out evaluation was repeated using an alternative 12-month test window. The ranking of models is broadly consistent with the main results: seasonal benchmarks remain competitive, while models that explicitly capture serial dependence and multiplicative seasonality (ETS/SARIMA) generally outperform additive baselines when volatility is moderate. This robustness check reinforces the conclusion that flexible seasonal models provide more reliable forecasts under varying market conditions.

Table 3: Test-set accuracy (12 months) starting 2018-01

Model	RMSE	MAE	MAPE	MASE
ETS (M,*,M)	16.84	14.66	35.20	1.23
SARIMA (auto)	18.57	15.56	34.37	1.30
Seasonal Naive	18.68	15.89	39.07	1.33

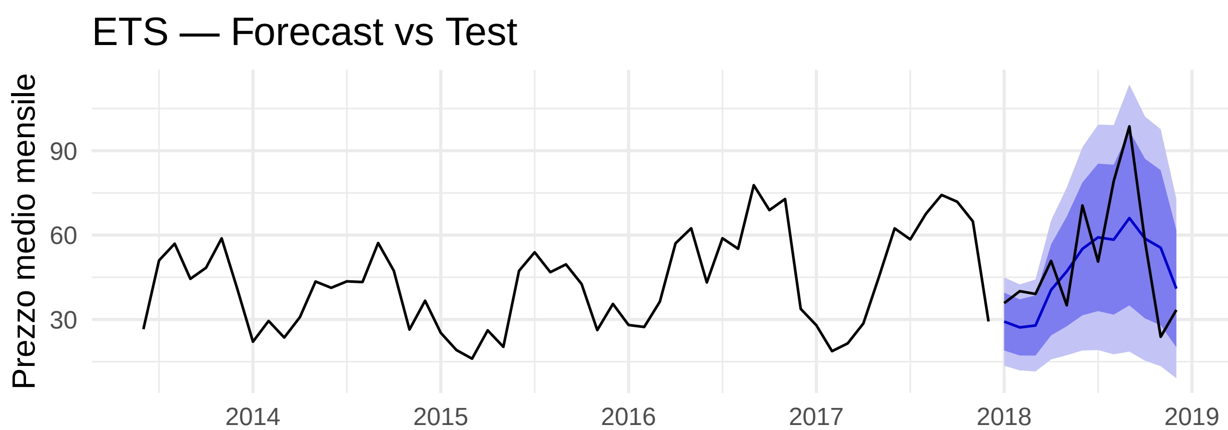


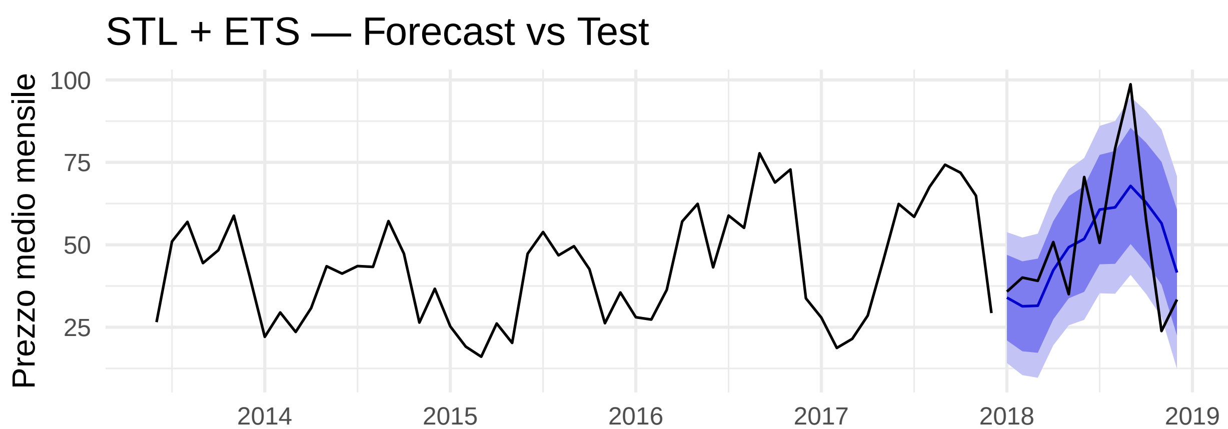
The figure compares ETS and SARIMA forecasts against the actual series in 2018. Both models track the upward trend, but their behaviors differ: ETS (dashed line) adapts more closely to the magnitude of seasonal fluctuations, resulting in lower absolute errors (RMSE = 16.8, MAE = 14.7). SARIMA (solid line), by contrast, smooths the trajectory, capturing the trend but underestimating peaks, which leads to slightly higher absolute errors but a marginally lower percentage error (MAPE = 34.4 vs 35.2). In any case, both models clearly outperform the seasonal naïve benchmark (MAPE = 39.1).

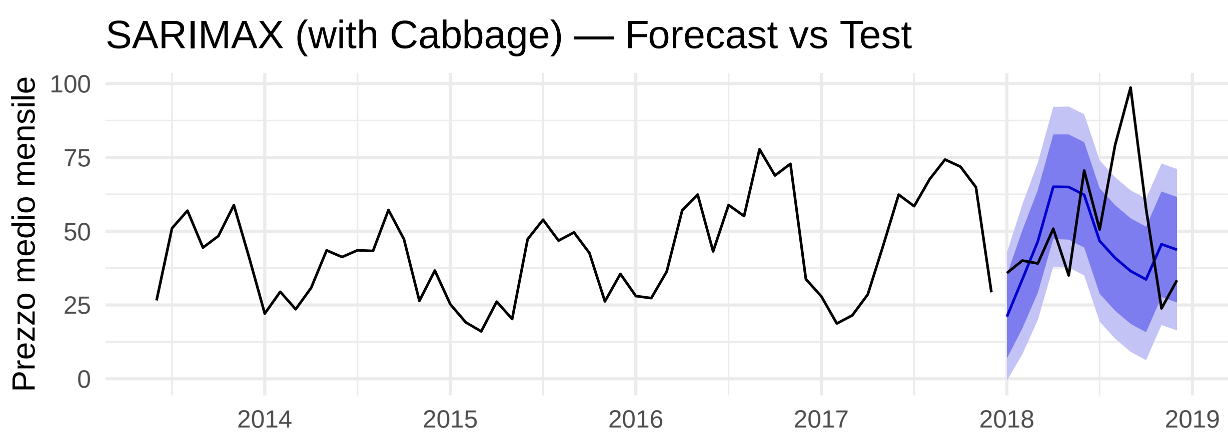
This experiment highlights that, under less volatile conditions, multiplicative ETS is more competitive in terms of absolute accuracy, whereas SARIMA provides a smoother forecast that may yield advantages in relative terms. The consistency of results across different test windows strengthens the robustness of the conclusions.

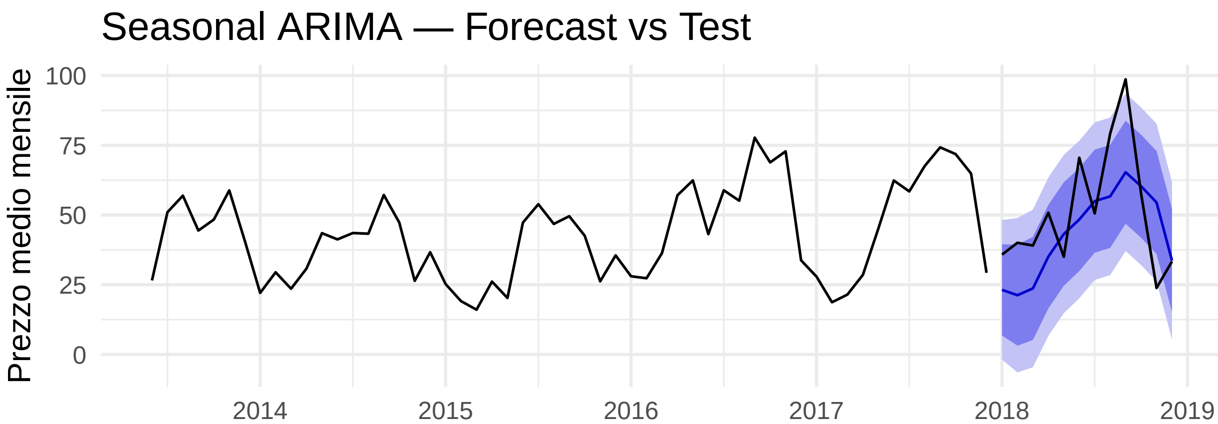
3.4 Other Models

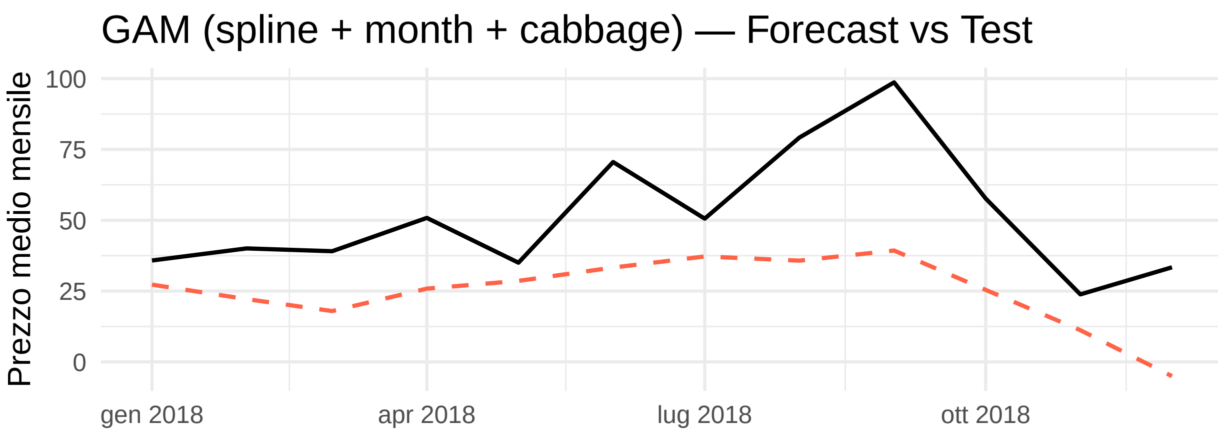
In addition to the baseline regressions, a set of alternative seasonal models was estimated, including ETS, STL+ETS, SARIMAX (with cabbage as exogenous regressor), and two variants of GAM (linear and log-transformed with cyclic seasonality). The following plots illustrate the forecasts of each model compared to the test period. A systematic comparison of their predictive accuracy is reported in the next section.

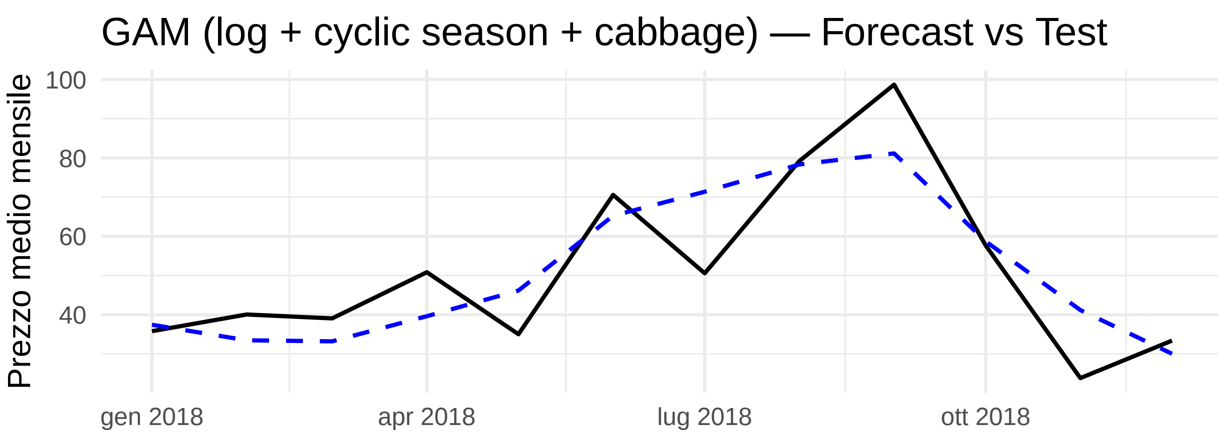

ETS — Forecast vs Test
Prezzo medio mensile
90
60
30
2014
2015
2016
2017
2018
2019

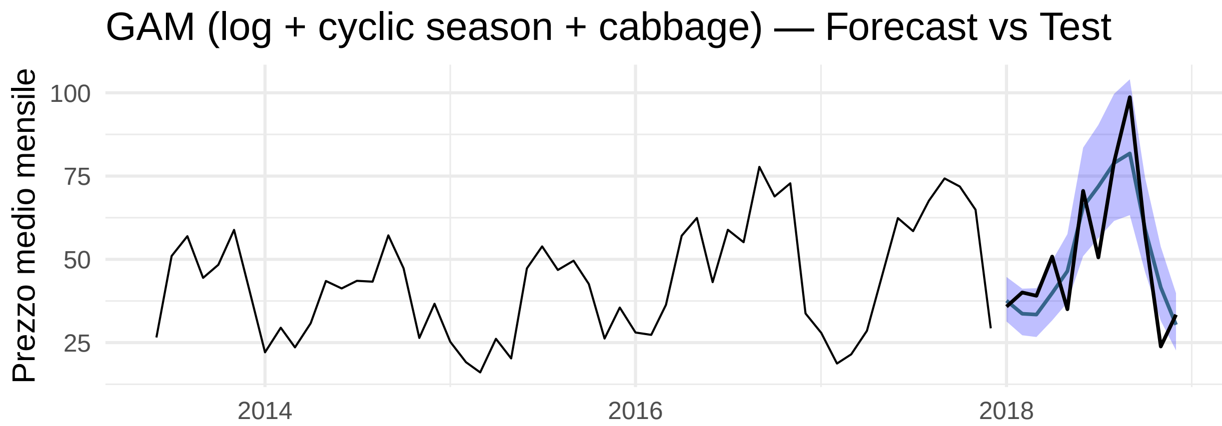

STL + ETS — Forecast vs Test
Prezzo medio mensile
100
75
50
25
2014
2015
2016
2017
2018
2019


SARIMAX (with Cabbage) — Forecast vs Test
Prezzo medio mensile
100
75
50
25
0
2014
2015
2016
2017
2018
2019


Seasonal ARIMA — Forecast vs Test
Prezzo medio mensile
100
75
50
25
0
2014
2015
2016
2017
2018
2019


GAM (spline + month + cabbage) — Forecast vs Test
Prezzo medio mensile
100
75
50
25
0
gen 2018
apr 2018
lug 2018
ott 2018


GAM (log + cyclic season + cabbage) — Forecast vs Test
Prezzo medio mensile
100
80
60
40
gen 2018
apr 2018
lug 2018
ott 2018


GAM (log + cyclic season + cabbage) — Forecast vs Test
Prezzo medio mensile
100
75
50
25
2014
2016
2018

4 Compare Different Models

Table 4: Forecasting model comparison (sorted by MAPE)

	Model	MAE	MAPE
9	GAM (log + cyclic season + cabbage)	8.556	20.168
6	STL + ETS (stlm)	13.705	31.222
5	ETS	14.241	32.483
3	TSLM: + Cabb+Pot+Onion	13.995	33.331
2	TSLM: + Cabbage	15.012	34.018
4	ARIMA (seasonal)	15.563	34.365
7	SARIMAX (+ Cabbage)	20.116	40.355
1	TSLM: trend+season	18.165	42.056
8	GAM (spline t + month + cab)	26.304	50.693

The analysis was expanded by estimating several seasonal forecasting models on the training sample and validating them on a 12-month hold-out window (2018). The comparison (Table 4) shows a clear ranking of model performance:

- The **log-GAM with cyclic seasonality and cabbage** achieves the best accuracy overall (MAE 8.6, MAPE 20%), reducing the error by more than half compared to the baseline TSLM. The logarithmic transformation effectively stabilizes variance, while the cyclic spline enforces continuity between December and January. Including cabbage as an exogenous regressor provides additional market information and reduces bias.
- Among purely univariate models, **STL+ETS** performs best (MAPE 31%), slightly ahead of standard ETS (MAPE 32%). Both methods adapt to changes in seasonal amplitude, which is crucial for agricultural series. **Seasonal ARIMA** also provides a reasonable fit (MAPE 34%), though less competitive than the exponential smoothing family.
- **SARIMAX (with cabbage)** performs worse (MAPE 40%). While the exogenous regressor is significant in-sample, its predictive contribution is unstable across years, and such models require reliable forecasts of future regressors, limiting their practical robustness.
- The **linear-scale GAM** is most affected by variance growth and records the highest errors (MAPE 50%).

Overall, models with multiplicative or log-based structures (ETS, STL+ETS, log-GAM) are better suited to capturing both trend and seasonal fluctuations. For transparent univariate forecasting, STL+ETS is a strong candidate; however, when reliable exogenous information is available, the log-GAM clearly provides the most accurate and robust forecasts.

5 Log Transform Effect

Table 5: Accuracy of models fitted on log-transformed series (back-transformed to original scale)

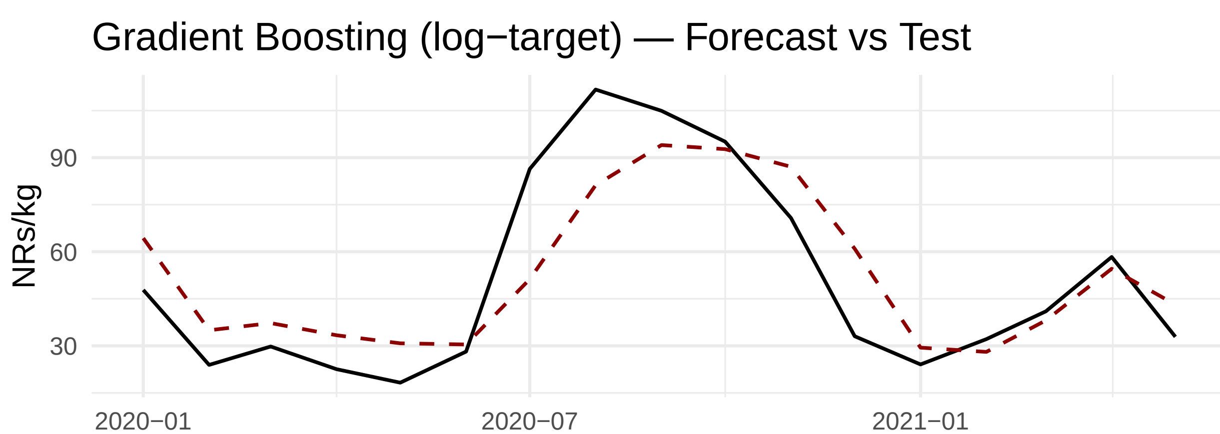
Model	ME	RMSE	MAE	MAPE
ETS (log)	-21.07	29.74	22.12	35.90
STL+ETS (log)	-5.12	16.65	14.43	31.61
SARIMA (log)	-9.94	18.87	15.34	31.93

As a final step, the main time series models were re-estimated on the log-transformed series, with forecasts back-transformed to the original scale for comparability. The log transformation aims to stabilize variance and capture relative (percentage) rather than absolute fluctuations in prices, which is particularly relevant for agricultural commodities.

The results confirm that the transformation can improve forecast performance in models sensitive to variance heterogeneity. In particular, STL+ETS on log-transformed data achieved the lowest out-of-sample error (MAE 14.4, MAPE 31.6%), followed closely by SARIMA (log) (MAE 15.3, MAPE 31.9%). By contrast, ETS (log) deteriorated substantially (MAE 22.1, MAPE 35.9%), suggesting that its multiplicative seasonal structure already captured variance scaling reasonably well on the original scale.

Finally, *Gradient Boosting* was also tested with engineered features and a log-transformed target. Its performance (MAPE 29%) remained below the best statistical alternatives. This outcome is not surprising given the relatively short length of the monthly series and the strong seasonal structure of agricultural prices, which classical models such as ETS, STL+ETS, and log-GAM capture more directly. Gradient Boosting tends to require larger datasets and richer exogenous information to fully exploit its flexibility; in small samples it may overfit noise rather than seasonal patterns.

Overall, while log transformation is not universally beneficial, it enhances robustness for models like STL+ETS and SARIMA, which flexibly accommodate evolving seasonality and autocorrelation. The improvement is modest in absolute terms, but it reinforces the importance of testing variance-stabilizing transformations in volatile commodity markets. Including a machine learning benchmark also highlights that more complex models do not always outperform well-specified statistical approaches in short and highly seasonal series.



6 Conclusion

This study explored the forecasting of cauliflower prices in the Kalimati wholesale market using a range of time series approaches. Starting from a transparent linear baseline, seasonal structures, exogenous regressors, and variance-stabilizing transformations were progressively introduced.

The results show that models with multiplicative or log-based seasonality perform substantially better than additive baselines. Among univariate approaches, STL+ETS consistently delivered robust forecasts with reasonable accuracy (MAPE 31%), slightly ahead of standard ETS and seasonal ARIMA.

Including exogenous regressors (e.g., cabbage prices) provided some in-sample gains but produced unstable out-of-sample performance, highlighting the practical difficulty of relying on contemporaneous or forecasted covariates. By contrast, the log-GAM with cyclic seasonality achieved the best accuracy overall (MAPE 20%), benefiting from both variance stabilization and flexible modeling of the within-year cycle.

Experiments with log-transformed series confirmed that the effect of variance stabilization depends on the underlying model: it improved STL+ETS and SARIMA, but not ETS.

Overall, the analysis demonstrates that forecasting in volatile agricultural markets requires models capable of handling strong and evolving seasonality, heteroskedasticity, and occasional shocks. For operational use, STL+ETS provides a reliable univariate baseline, while the log-GAM emerges as the most accurate option when reliable exogenous information can be incorporated.